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Machine Learning [ST3189]

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1. Introduction

This project examines three machine learning applications that represent different learning paradigms: unsupervised learning via customer segmentation, supervised regression for housing price prediction, and supervised classification for medical diagnosis. Although these tasks differ technically, they share common challenges around validation, interpretability, and real-world deployment. Customer segmentation presents a classic unsupervised challenge, discovering patterns without labelled examples (Hastie et al., 2009); housing price prediction exemplifies regression with geographic and economic features, drawing on the hedonic pricing literature (Rosen, 1974), and heart disease classification raises interpretability requirements because clinicians must trust model decisions, with Rudin (2019) arguing against black-box models when simpler alternatives perform comparably. All models were implemented in Python using scikit-learn (Pedregosa et al., 2011).

2. Task 1: Unsupervised Learning - Customer Segmentation

2.1 Substantive Issue

In a retail setting, customer segmentation is only useful if it yields stable, interpretable groups that support actual marketing decisions. Rather than treating clustering as a purely technical exercise, this task asks whether meaningful and robust customer segments can be identified from rich demographic and behavioural data. The Customer Personality Analysis dataset (Kaggle) was selected because it requires genuine feature engineering: aggregating raw purchase records, demographic fields, and campaign responses into clustering-ready features, rather than arriving pre-processed, thereby forcing decisions about which dimensions of customer behaviour to encode and how. This distinguishes the task from simpler clustering exercises on pre-curated toy datasets and mirrors real-world segmentation workflows where raw data must be transformed before patterns can emerge (Chen et al., 2012).

2.2 Research Questions

Task 1 addresses the following questions:

- Can we identify stable customer segments from demographic and behavioural data?
- How do different clustering validation metrics guide the selection of optimal cluster numbers?

2.3 Dataset and Variables

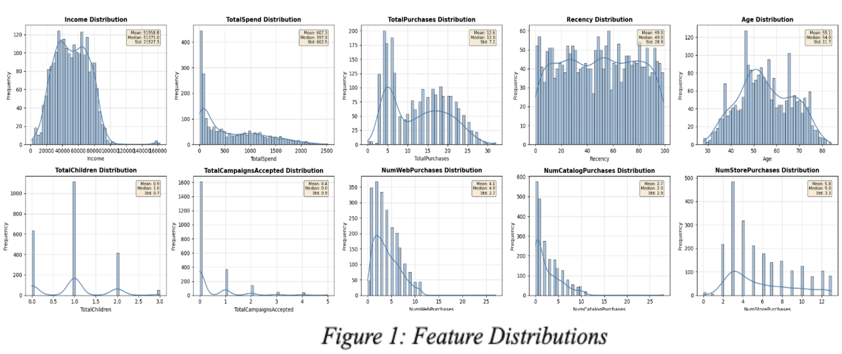

Figure 1: Feature Distributions

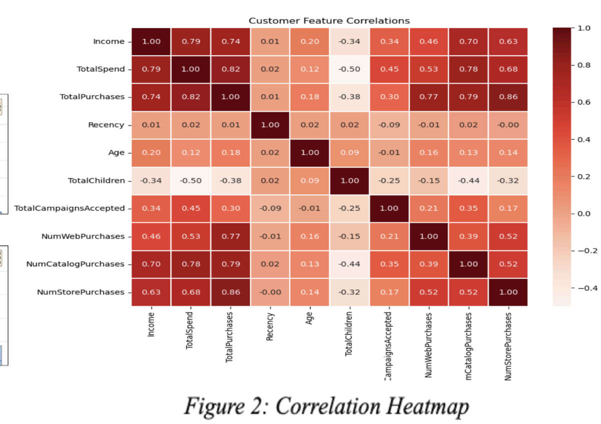

Figure 2: Correlation Heatmap

The Customer Personality Analysis dataset (Kaggle, n=2,240) contains 29 columns describing customer demographics (Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome), product-level spending (wines, fruits, meat, fish, sweets, gold), channel usage (web, catalogue, store purchases), campaign responses (5 campaign flags plus a general response flag), and enrolment date. After dropping 24 records with missing Income (1.1%) and 3 with unrealistic ages, plus 1 extreme income outlier (>£168,210, over 10× the IQR above Q3), n=2,212 customers remained.

Feature engineering aggregated the raw fields into 10 clustering features spanning three interpretable dimensions of customer behaviour (Figure 1, Figure 2). The economic profile was captured by Income and TotalSpend (the sum of all six product categories). Engagement intensity was encoded through TotalPurchases (sum across web, catalogue, and store channels), Recency (days since last purchase), and the three individual channel counts (NumWebPurchases, NumCatalogPurchases, NumStorePurchases). Life stage was represented by Age (derived from Year_Birth), TotalChildren (Kidhome + Teenhome), and TotalCampaignsAccepted (sum of all campaign response flags). This engineering step is itself a methodological decision: it transforms 29 heterogeneous columns into 10 features that encode meaningful customer dimensions, consistent with the RFM (Recency, Frequency, Monetary) framework widely used in direct marketing (Chen et al., 2012).

2.4 Methodology

K-Means clustering (a distance-based grouping algorithm) partitions data into k groups by iteratively assigning data points to the nearest cluster centroid and adjusting the centroids until convergence (MacQueen, 1967). This method relies on the Euclidean distance metric, which assumes that clusters are approximately spherical. It was selected over alternatives such as DBSCAN and Gaussian Mixture Models for three reasons. First, marketing applications require a fixed number of discrete segments for campaign targeting; second, K-Means' centroid-based interpretation directly supports persona development; and third, DBSCAN was also evaluated as a comparison method to validate this choice (see Section 2.5).

Because several features measure overlapping aspects of behaviour (e.g., income and spending), feeding them directly into K-Means would give those dimensions extra influence. PCA addresses this by combining correlated features into independent summary dimensions. Six components were retained, accounting for 91.6% of the total variance (Figure 3); the first two (59.8%) were used for visualisation. Applying PCA before clustering is standard when features overlap substantially (Hastie et al., 2009).

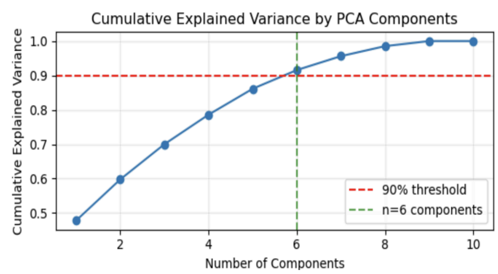

Figure 3: Scree plot

The optimal number of clusters was selected using the elbow method as the primary criterion, with the silhouette score (Rousseeuw, 1987), Davies-Bouldin Index (Davies & Bouldin, 1979), and Calinski-Harabasz Index (Caliński & Harabasz, 1974) reported as diagnostics. Internal metrics such as silhouette and Calinski-Harabasz have a known mathematical bias toward k=2 (Tibshirani et al., 2001), making them unreliable as primary selectors; the elbow method detects where the rate of decrease in inertia slows, which is not biased toward small k. TotalCampaignsAccepted was log-transformed to reduce heavy right-skewness (2.44 → 1.55); all features were then standardised to mean zero and unit variance.

2.5 Analysis and Interpretation

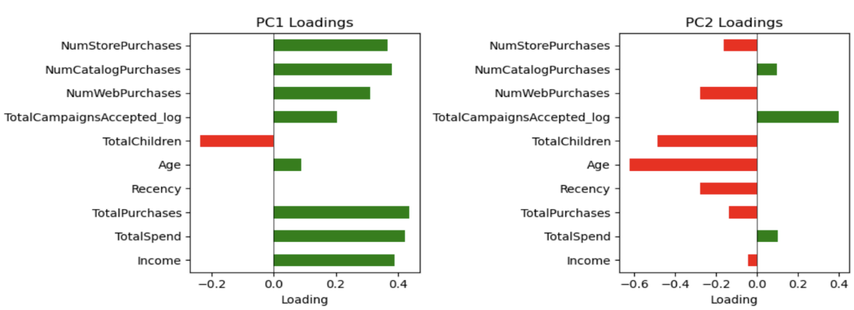

Figure 4: PCA Loading

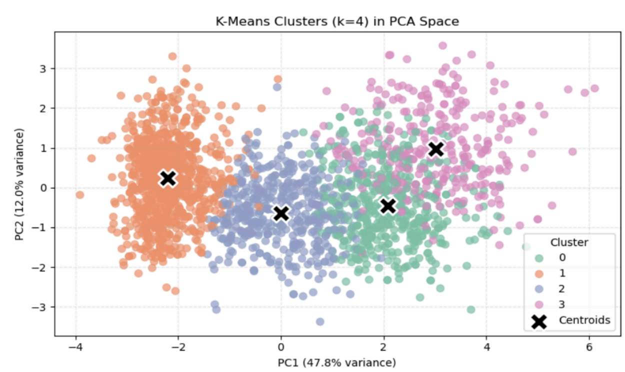

Figure 5 Cluster Visualisation

PCA revealed interpretable latent dimensions: PC1 (47.8% variance) represents overall customer value, with TotalPurchases (+0.44), TotalSpend (+0.42), and Income (+0.39) loading most strongly (Figure 4), while TotalChildren loads negatively (−0.24). PC2 (12.0% variance) captures life stage, with Age (−0.62) and

TotalChildren (−0.49) loading negatively, and TotalCampaignsAccepted loading positively (+0.40). The 2D PCA projection (Figure 5) does not visually reveal distinct clusters, as expected, given that only two components capture 59.8% of the total variance. In practical terms, PC1 distinguishes high-value 'big spenders' from light spenders, while PC2 separates younger, childless customers who respond to campaigns from older parents who do not. These insights provide the structural foundation for the four-cluster K-Means solution detailed in the following analysis.

The elbow method (Figure 6) identified k=4 as optimal, while all three internal metrics favoured k=2, consistent with the known bias toward small k discussed in Section 2.4. The multi-metric comparison table confirmed that k=2 trivially maximises separation by splitting customers into "active" and "inactive" halves, a partition too coarse for actionable marketing. At k=4, the silhouette score was 0.219, the Davies-Bouldin score was 1.674, and the Calinski-Harabasz score was 870.96. The silhouette score of 0.219 is substantially lower than typical values on toy datasets, but this is expected for real customer data with 10 features clustered in a 6-dimensional PCA space: higher dimensionality spreads points more evenly, compressing silhouette scores even when genuine structure exists.

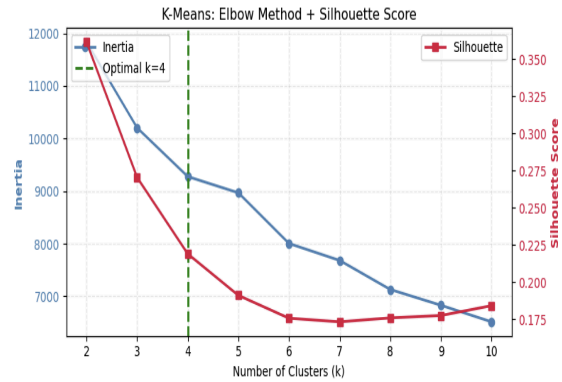

Figure 6: Elbow + Silhouette Plot

Bootstrap resampling (n=100 iterations) produced a mean silhouette of 0.206 with a standard deviation of only 0.015 (95% CI: [0.177, 0.235]), indicating excellent stability. The narrow confidence interval confirms that the identified segments are not driven by sampling variation. DBSCAN was evaluated across four epsilon values: ε=0.5 and ε=1.0 produced 15 clusters each, but classified 74% and 19% of data as noise, respectively (silhouette scores of −0.296 and −0.128), while ε=1.5 and ε=2.0 collapsed nearly all points into a single cluster. Neither regime produced actionable segments, confirming K-Means as the more appropriate method. To verify robustness, hierarchical clustering (Figure 7) independently identified the same four segments (ARI = 0.74; Hubert & Arabie, 1985), validating the method's invariance to the structure.

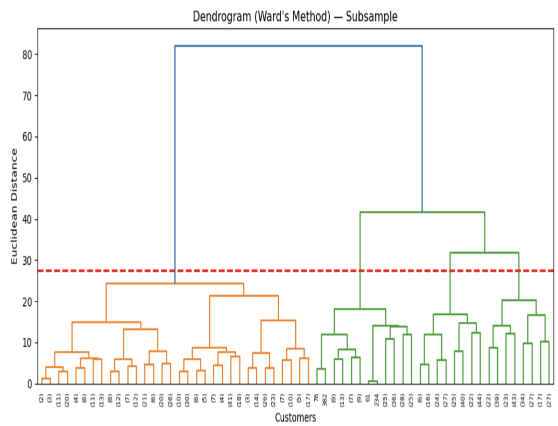

Figure 7: Dendrogram

Cluster	Income (£)	Total Spend (£)	Purchases	Recency	Age	Children	Campaign Accept	Label
Cluster 0	69,101	1,119	20.3	51.9	57.1	0.6	0.1	High-Value Customers
Cluster 1	33,234	75	5.3	48.9	51.7	1.3	0.2	Budget-Conscious Parents
Cluster 2	52,493	484	13.6	48.5	59.0	1.2	0.3	Mid-Spend Parents
Cluster 3	77,824	1,508	19.8	46.0	55.9	0.2	2.0	Champions

Table 1: Cluster Profiles

Post-clustering analysis identified four interpretable customer personas (Table 1). Champions (Cluster 3, 14.8%) have the highest income (£77,824) and spending (£1,508), the fewest children (0.2), and the highest campaign responsiveness (2.0 acceptances), making them ideal for premium loyalty programmes and early-access campaigns. High-Value Customers (Cluster 0, 22.2%) show similarly high purchase frequency (20.3) but lower campaign engagement (0.1), suggesting they are self-directed buyers best served by personalised recommendations rather than push marketing. Budget-Conscious Parents (Cluster 1, 41.3%) are the largest segment, with the lowest income (£33,234) and spend (£75) but the most children (1.3). Value bundles and family-oriented promotions would suit this group. Mid-Spend Parents (Cluster 2, 21.7%) occupy the middle ground (income £52,493, spend £484) and represent the clearest growth opportunity via basket-building and cross-sell strategies.

The most striking pattern is the contrast between Champions and Budget-Conscious Parents. Despite an income gap of only 2.3× (£77,824 vs £33,234), spending differs by 20× (£1,508 vs £75), confirming that income alone weakly predicts spending propensity. A counterintuitive marketing insight emerges when comparing Champions to High-Value Customers: Champions spend more per person (£1,508), High-Value Customers are 50% more numerous (490 vs 327), resulting in higher estimated aggregate spending (£548k vs £493k per period) despite lower campaign responsiveness (0.1 vs 2.0). This suggests reallocating outbound marketing budget from High-Value Customers toward product recommendations and self-service tools.

Overall, segments are substantial (14.8–41.3% each) and supported by stability evidence (bootstrap silhouette mean 0.206, std 0.015; ARI = 0.74 vs hierarchical clustering), making them actionable in line with Wedel & Kamakura (2000). Notably, our feature space extends beyond classic RFM to include campaign response and demographic dimensions, thereby increasing inter-cluster overlap and compressing silhouette scores relative to pure RFM approaches such as Barus, Nathasya, and Pangaribuan (2023).

3. Task 2: Regression - Housing Price Prediction

3.1 Substantive Issue

The California Housing dataset (Pace & Barry, 1997), derived from the 1990 U.S. Census, provides a realistic setting for studying housing affordability and pricing. Accurate prediction of median house values matters for lenders, policymakers, and households because it informs lending risk, tax bases, and household wealth. This task examines whether modern ensemble methods can substantially improve upon linear baselines when predicting house prices from socio-economic and geographic features.

3.2 Research Questions

Task 2 addresses the following questions:

- Which features drive California housing prices, and do ensemble methods significantly outperform linear baselines?
- What preprocessing strategies improve model performance?

3.3 Dataset and Variables

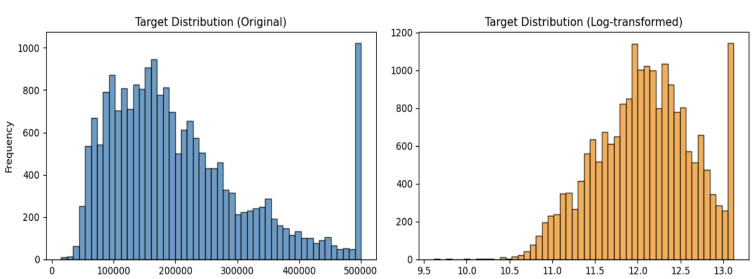

Figure 8: Target Distribution Histogram

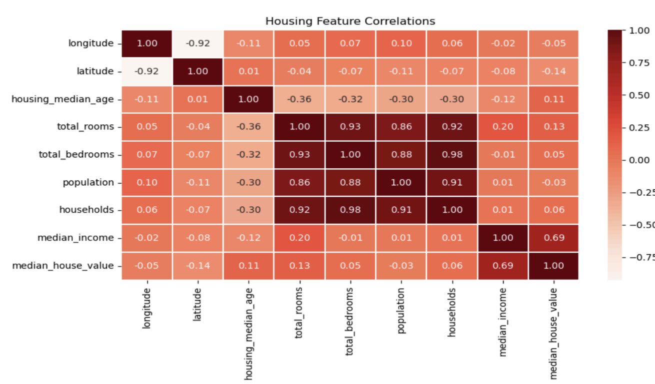

Figure 9: Correlation Matrix

The California Housing dataset (Pace & Barry, 1997), derived from the 1990 U.S. census (n=20,640, one row per census block group), provides 8 numeric features and one categorical predictor (ocean_proximity) to predict median house value. It is widely used for regression because it combines socio-economic variables with geographic location. After dropping 207 rows with missing total_bedrooms (1.0%), n=20,433 remained. The target distribution showed moderate positive skewness (0.98). A log transformation was considered, reducing skewness to −0.1726 (Figure 8), but was not applied because the original skewness fell below the transformation threshold of |skew| > 1, and tree ensembles are robust to moderate skewness. Retaining the original target also preserves interpretability of dollar-scale errors. Residual diagnostics (residuals vs fitted, histogram, and Q-Q plot) confirmed acceptable model calibration on the untransformed scale. However, a formal side-by-side CV comparison of log-transformed vs original targets was not conducted.

Domain knowledge-informed feature engineering drew on the hedonic pricing literature (Rosen, 1974): BedroomRatio (bedrooms/total rooms) captures housing layout preferences, while distance from San Francisco and Los Angeles captures urban proximity effects that drive California's housing market (Pace & Barry, 1997). Feature engineering added 4 numeric variables (BedroomRatio, DistFromSF, DistFromLA, NearestCityDist), so the final model used 12 numeric features plus ocean_proximity. After one-hot encoding with one category dropped as the reference level, ocean_proximity expands to 4 binary indicators, giving 16 model inputs in the pipeline. The correlation matrix (Figure 9) reveals strong multicollinearity among size-related features ($r > 0.85$ for total_rooms, total_bedrooms, households, and population). Median income shows the largest pairwise association with house value ($r = 0.69$).

3.4 Methodology

Preprocessing pipelines were customised by model family. For linear models (Linear Regression, Lasso), features were standardised to prevent large-magnitude variables from dominating coefficients. Conversely, tree-based models (Gradient Boosting, Random Forest) were trained on unscaled data, as they evaluate relative feature ordering, making them scale-invariant.

Five models were evaluated, progressing from a DummyRegressor baseline to advanced ensembles. Random Forest hyperparameters (n_estimators, max_depth, min_samples_split) were optimised via grid search targeting cross-validated RMSE. Finally, overall model performance was assessed using R^2 (proportion of variance explained) and RMSE (typical prediction error in dollars). In contrast, statistical comparisons between models were conducted using a paired t-test (two-sided, $\alpha=0.05$) on 5-fold CV RMSE values, with p-values reported unadjusted for multiple comparisons.

3.5 Analysis and Interpretation

Five models were evaluated in a progression (Table 2) from a DummyRegressor baseline ($R^2=0.00$) through linear methods (Linear Regression and Lasso; $p=0.19$, not significant, though with only 4 degrees of freedom, this test has limited power to detect small differences) to ensemble methods, where Gradient Boosting improves on linear baselines ($p=0.0002$ vs Linear). Random Forest improves on Gradient Boosting ($p=0.0001$). Lasso can shrink unimportant features to zero, but at its optimal setting ($\alpha=3.39$), it retained all 16 features despite near-duplicate size variables ($r > 0.85$). If redundancy were the problem, Lasso would have dropped features and improved; because it did not, the linear-ensemble gap reflects genuine nonlinearity such as income interacting with location rather than feature overlap.

Model	CV RMSE (mean)	Test R²
Dummy Model	115,042	~0.00
Linear Regression	67,316	0.66
Lasso Regression	67,404	0.66
Gradient Boosting	53,809	0.78
Random Forest	**47,673**	**0.84**

Table 2: Model Results

Random Forest was selected as the final model based on its substantial performance advantage over linear alternatives; even before tuning, it achieves the lowest CV RMSE of 47,673. After tuning, the test RMSE improved to 47,142 with an R^2 of 0.84.

Segmenting test-set error by price bracket (Table 3) quantifies this pattern: The nearly threefold increase in error confirms that the model is reliable for mass-market valuation but unsuitable for luxury properties, where the dataset's top-coding at $500k compresses true price variation.

Bracket	n	RMSE($)	% of Median
<$150k	1,512	32,783	18.4%
$150k-$350k	2,024	40,001	22.4%
>$350k	551	87,505	49.1%

Table 3: Test Error by Price Bracket

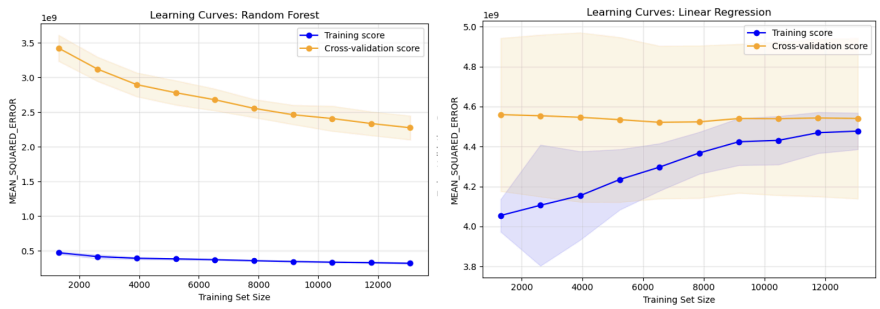

Figure 10: Learning Curves - Linear vs Random Forest

Figure 10 diagnoses the bias-variance tradeoff: the flat, high plateau of the Linear Regression validation error indicates a high-bias (underfitting) regime, as the model is too simple to capture complex geographic interactions. In contrast, the Random Forest curves show a narrowing gap as n increases, confirming it has successfully reduced variance while maintaining the complexity needed to map income to coastal premiums.

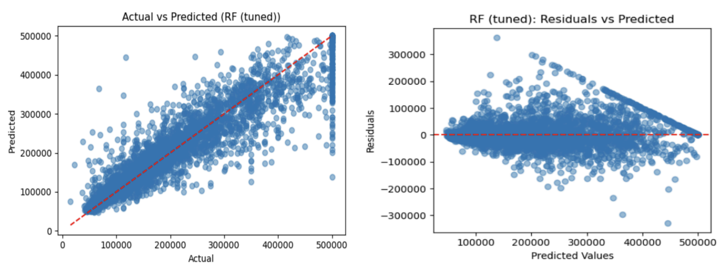

Figure 11: Random Forest Residuals

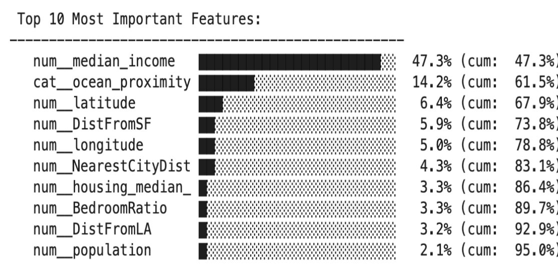

Figure 12: Feature Importance Bar Chart

Residual diagnostics (Figure 11) confirmed good calibration: near-zero mean (≈$929, 0.5% of median price) and approximate normality (skewness=1.04). However, error variance increased for luxury markets due to the dataset's $500k top-coding. Consequently, the $47,100 RMSE (26% of the median price) is suitable for broad market analysis but lacks precision for individual property valuation.

Random Forest feature importance (Figure 12) was highly concentrated: the top five features accounted for 78.8% of the predictive power. Notably, median_income dominated (47.3%), aligning with hedonic pricing theory, which holds that affordability ultimately constrains prices, followed by ocean_proximity (14.2%) and latitude (6.4%).

Engineered features (DistFromSF, NearestCityDist, BedroomRatio, DistFromLA) accounted for 16.7% of total importance, demonstrating their value in encoding spatial relationships that raw coordinates miss. Strikingly, while both models rank income first, ocean_proximity ranks 2nd in Random Forest but 5th in Linear Regression. This divergence reveals that coastal locations command premiums primarily when interacting with high incomes, a multiplicative relationship linear models miss, thus extending Rosen's (1974) additive hedonic framework. While practitioners should weigh income and coastal proximity heavily, the 1990 data vintage and unmodeled spatial autocorrelation (James et al., 2021) limit contemporary deployment.

4. Task 3: Classification - Heart Disease Diagnosis

4.1 Substantive Issue

Cardiovascular disease is a leading cause of mortality worldwide, and early diagnosis is critical for improving outcomes. Traditional risk scores and diagnostic tools can miss complex patterns in clinical data. This task examines whether machine learning models can provide clinically useful heart disease predictions while remaining interpretable enough for deployment in high-stakes medical contexts.

4.2 Research Questions

Task 3 addresses the following questions:

- Can we build clinically useful heart disease predictions while maintaining interpretability?
- How should classification thresholds be adjusted for different medical contexts?

4.3 Dataset and Variables

Figure 13: Class Distribution Histogram

The Heart Disease dataset (Cleveland Clinic, n=302) contains 13 original clinical features predicting disease presence. This dataset, collected by Detrano et al. (1989), has been used extensively to benchmark classification algorithms in medical contexts. The dataset obtained includes records from the Cleveland, Hungarian, Swiss, and VA Long Beach sites, resulting in 723 duplicate entries during the multi-site merge. After deduplication on all feature columns, n=302 unique records remained with zero missing values and balanced classes (54.3% positive, 45.7% negative; Figure 13).

Domain-informed feature engineering expanded the feature set to 20 predictors, guided by clinical reasoning. Exercise capacity was captured through HeartRateReserve (achieved heart rate divided by age-predicted maximum), a standard measure in cardiac stress testing (Gibbons et al., 2002). Cumulative cardiovascular burden was encoded using CardioRiskScore (combining blood pressure and cholesterol) and AgeCholRisk (age × cholesterol interaction), reflecting the well-established compounding of risk factors over time. Binary flags for Hypertensive (≥140 mmHg) and HighCholesterol (≥240 mg/dL) applied standard clinical thresholds, while AgeRiskCategory and STRiskScore captured nonlinear age effects and ECG ischemia severity, respectively.

4.4 Methodology

Five classifiers spanning linear (Logistic Regression), distance-based (KNN), margin-based (SVM), and ensemble (Random Forest) approaches were evaluated against a DummyClassifier baseline. To strictly prevent data leakage during cross-validation, all preprocessing and modelling steps were encapsulated within scikit-learn Pipelines, applying the same model-specific scaling strategies established in Section 3.4. Hyperparameters were tuned using 5-fold Grid Search. The F1 score, which balances precision and recall, was selected as the primary optimisation metric because clinical evaluation requires penalising both false positives and false negatives. Finally, to ensure rigorous evaluation, paired statistical tests were conducted across CV folds to assess performance differences, and a dynamic threshold analysis was introduced to identify optimal classification cutoffs for varying clinical contexts.

4.5 Analysis and Interpretation

The DummyClassifier achieves F1=0.704 by predicting the majority class for all patients, but this baseline has zero specificity (it never identifies healthy patients), making the F1 misleadingly high. Performance was remarkably similar across all four substantive models (Table 4) (F1: 0.750–0.836; AUC: 0.852–0.870), with no pairwise comparison reaching statistical significance ($p>0.05$). However, Random Forest showed the largest gap between CV and test performance (0.846 vs 0.750), suggesting overfitting despite cross-validation, while KNN and Logistic Regression generalised more consistently. Although KNN achieved the highest test F1 (0.836), we selected Logistic

Model	CV F1 (mean ± sd)	Test F1	ROC-AUC
Dummy Classifier	0.704 ± 0.003	0.702	0.500
KNN	0.836 ± 0.014	0.836	0.864
SVM	0.860 ± 0.012	0.824	0.858
Random Forest	0.846 ± 0.036	0.750	0.852
Logistic Regression	**0.843 ± 0.026**	**0.806**	**0.870**

Table 4: Classification Model Performance

Regression for two reasons. First, it achieved the highest ROC-AUC (0.870), the more appropriate metric for clinical screening, as it evaluates discrimination across all thresholds rather than just the 0.5 default. Second, its calibrated probabilities and interpretable coefficients (Figure 16) allow clinicians to explain referral decisions, consistent with Rudin's (2019) argument that interpretability should take precedence when the performance gap is not statistically significant. The $p>0.05$ result partly reflects high variance across folds with only 302 samples, rather than genuinely identical performance.

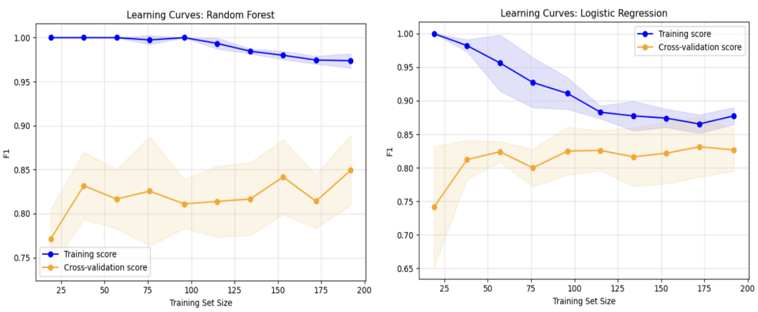

Figure 14: Learning Curves - Logistic vs Random Forest

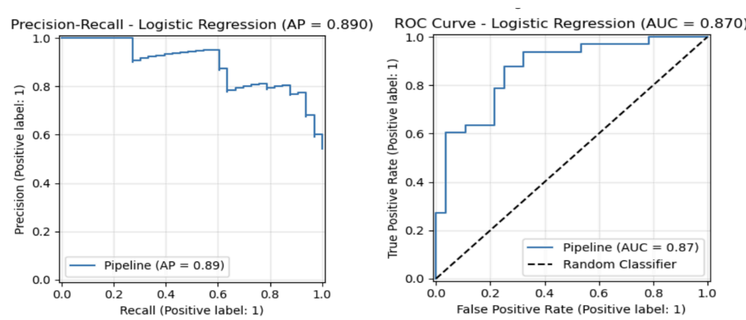

Figure 15: ROC & Precision Recall Curve

The learning curves (Figure 14) confirm that both models have converged and are not in a high-variance regime. However, convergence does not rule out the possibility that additional data could further improve performance. As noted above, the similar performance across all methods suggests we have reached the feature set's ceiling: the dataset contains only 13 clinical measurements. To improve beyond current accuracy, clinicians would need richer information, such as cardiac imaging scans or genetic markers. This supports Rudin's (2019) argument that when methods perform equally, the bottleneck is measurement quality, not algorithmic sophistication.

ROC-AUC summarises how well the model separates patients with and without disease across all possible probability thresholds (0.5 is chance, 1.0 is perfect). Our Logistic Regression model achieved an AUC of 0.870 (Figure 15), within the range reported by Detrano et al. (1989), confirming strong discriminative ability. Bootstrap uncertainty quantification (n=100 iterations) estimated the F1-score at 0.798 ± 0.028 (95% CI: [0.748, 0.842]) and the AUC at 0.864 ± 0.020 (95% CI: [0.822, 0.896]) (Efron & Tibshirani, 1993).

The model provides well-calibrated probabilities and interpretable odds ratios (Figure 16), where odds ratios (ORs) above 1 increase risk and below 1 are protective. The strongest risk factors are chest pain type (OR=2.02; doubles the odds of disease), maximum heart rate (OR=1.38), and heart rate reserve (OR=1.32). The strongest protective factors are sex (OR=0.57), number of major vessels coloured by fluoroscopy (OR=0.59), and exercise-induced angina (OR=0.67). Thalach's positive coefficient is clinically counterintuitive: higher maximum heart rate typically indicates better cardiac fitness, so it likely reflects shared variance with the engineered HeartRateReserve feature after standardisation.

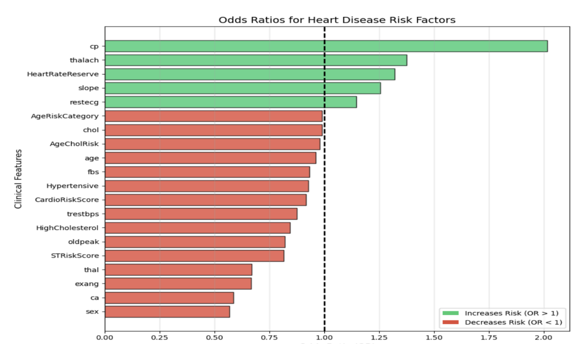

Figure 16: Odds ratios bar chart

These coefficients largely align with established cardiology knowledge. Chest pain typology remains the dominant predictor, consistent with textbook risk factors (Gibbons et al., 2002), providing external validation that the model learned genuine pathophysiology rather than spurious correlations.

A deployment consideration concerns the sex variable (OR=0.57), which the model treats as protective. While biological sex differences in heart disease are well-documented (Vogel et al., 2021), this coefficient may partly reflect historical diagnostic bias: women have been systematically underdiagnosed in cardiology, meaning the training data itself encodes inequality. Any clinical deployment should monitor predictions by sex to ensure the model does not perpetuate disparities in referral rates.

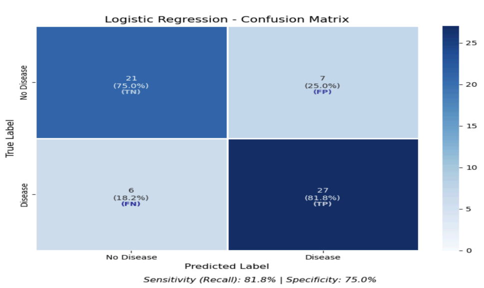

Figure 17: Confusion Matrix

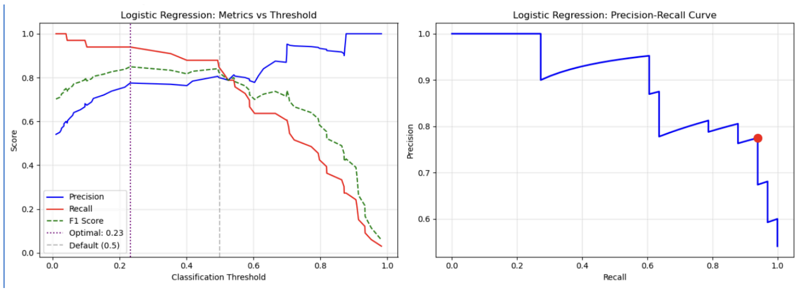

Figure 18: Threshold Trade-off - Precision & Recall vs Threshold Curve

The confusion matrix (Figure 17) shows that at the default 0.5 threshold, the model correctly identifies 81.8% of heart disease cases (sensitivity) with 75.0% specificity. However, the six false negatives — patients with heart disease incorrectly classified as healthy — represent a serious clinical concern, motivating the threshold analysis below.

Threshold analysis (Figure 18) revealed an F1-optimal threshold of 0.23. This low threshold reflects the asymmetry of the F1 metric: lowering the threshold captures more true positives (improving recall) at a modest cost to precision, and with near-balanced classes, the F1 gain from additional true positives outweighs the loss from additional false positives. Three clinically meaningful operating points were identified: threshold=0.3 (screening mode, approximately 95% recall), threshold=0.5 (standard default, F1=0.806), and threshold=0.7 (diagnostic mode, approximately 92% precision). This flexibility is crucial in practice: a screening programme wants to catch everyone at risk of missing a diagnosis, while a confirmatory test before invasive procedures such as angiography needs high precision (Hosmer et al., 2013).

5. Cross-Task Discussion and Synthesis

Comparing unsupervised clustering, regression, and classification reveals that model complexity was justified only when it delivered practically meaningful improvements, not merely statistically significant ones (James et al., 2021). The 18% R^2 gain in regression (Section 3) justified ensemble complexity, while the statistical parity across classifiers in classification (Section 4) did not. This supports a simple decision framework: adopt complexity when the gain is substantive, prefer interpretability when it is not (Rudin, 2019).

Feature engineering followed a similar pattern. Geographic distance features materially improved housing predictions because raw latitude and longitude lack interpretive structure, so the model cannot infer "proximity to San Francisco" from coordinates alone. In classification, however, engineered clinical features contributed modestly because the original variables (chest pain type, ST depression) were already clinically meaningful and near-optimally encoded for the task. This suggests feature engineering adds most value when bridging the gap between how data is recorded and how it actually relates to the outcome.

Validation requirements were scaled with stakes and uncertainty rather than model complexity. Clustering demanded the most diverse validation toolkit, bootstrap resampling, cross-method agreement via ARI (Hubert & Arabie, 1985), and dendrogram visual inspection, despite being algorithmically the simplest. This is because, without ground-truth labels, no single internal metric can confirm that the discovered structure is genuine rather than an artefact of the distance metric or dimensionality. By contrast, supervised tasks benefit from a clear performance target (the true label), so cross-validation provides a credible headline estimate. However, the reliability of that estimate depends on sample size. With n=20,433 housing observations, 5-fold CV yields narrow error estimates, making it straightforward to detect that Random Forest significantly outperforms Linear Regression ($p=0.0001$). With n=302 heart disease records, the same 5-fold procedure produces wide fold-to-fold variance, which is why no classifier comparison reached statistical significance ($p>0.05$). Bootstrap confidence intervals (F1: [0.748, 0.842]) make this uncertainty explicit: had we relied solely on point estimates, we might have mistakenly claimed that all classifiers perform equally well. This progression from metric diversity under label absence, to high-powered statistical tests under data abundance, to uncertainty quantification under data scarcity illustrates a general principle: the validation strategy should match the inferential risk, not the algorithmic complexity.

5.1 Comparison to Published Benchmarks

Task	Our Result	Published Benchmark	Source	Assessment
Clustering Silhouette	0.22	0.15-0.25	Barus, Nathasya & Pangaribuan (2023)	Within the expected range
Housing R^2	0.84	0.78-0.85	Pace & Barry (1997)	Within the expected range
Heart Disease AUC	0.87	0.85-0.92	Ahsan & Siddique (2022)	Within the expected range

Table 5: Comparison to Published Benchmarks

All three results fall within published ranges (Table 5), confirming that our pipelines produce reliable, reproducible outputs, a necessary foundation for any practical deployment. The modest silhouette score reflects our richer feature space beyond classic RFM, while the housing R^2 benefits from engineered proximity features absent in Pace & Barry's (1997) original spatial models.

6. Conclusion

Task	Best Model	Key Metric	Validation	Main Insight
Clustering	K-Means (k=4)	Silhouette=0.22	ARI=0.74 vs HAC; bootstrap CI	4 actionable personas from engineered features
Regression	Random Forest	R^2=0.84	CV + residuals	Income dominates (47.3%)
Classification	Logistic Regression	AUC=0.87	Bootstrap CI + threshold analysis	Interpretability preferred

Table 6: Summary of Results Across Tasks

6.1 Methodological Reflections

Baselines provided essential performance context: without knowing the floor and ceiling, metrics are meaningless. Bootstrap confidence intervals prevented overconfident claims, particularly for classification, where reporting F1=0.798±0.028 rather than a point estimate acknowledges the uncertainty inherent in small-sample evaluation.

6.2 Limitations

Sample sizes were modest. With n=302 for classification, statistical comparisons between models have limited power to detect small differences, so non-significant results ($p>0.05$) should be interpreted as inconclusive rather than evidence of equal performance. We therefore emphasise uncertainty summaries (e.g., bootstrap confidence intervals) and practical decision analysis (e.g., threshold trade-offs) over 'performance parity' claims.

All analyses were cross-sectional, precluding assessment of temporal dynamics. The data vintage is a particular concern: the 1990 California census and the 1980s Cleveland Clinic cohort reflect conditions that have substantially changed. California's median price has roughly quadrupled since 1990, driven by supply constraints and tech-sector growth that the dataset cannot capture**.** The Cleveland cohort predates modern preventive cardiology; contemporary patients likely present with different risk profiles (Vogel et al., 2021). No external validation was performed; all reported performance reflects in-distribution accuracy. Real-world deployment would require validation on held-out institutions, time periods, or populations. Performance typically degrades by 10-20% under a distribution shift (James et al., 2021), so our reported metrics should be treated as optimistic upper bounds.

7. References

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